Draft version October 25, 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

Rotation Periods and Spin Senses of Asteroids from Gaia DR3 Sparse Photometry, with Calibrated Reliability and Null Tests

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ABSTRACT

I fitted a periodogram to every numbered asteroid with at least 25 Gaia DR3 transits (41,842 solved). The model has season offsets, a linear phase term and a second-order Fourier series. The rotational phase is corrected for the phase-angle bisector (PAB), with both senses of rotation tried. The fit therefore returns the spin sense as well as the period.

Asteroids with a known period were split once into a tuning half and a test half. The period-doubling rule and a logistic model for the probability of a correct period were fitted on the tuning half. The confidence threshold was taken from two null tests that destroy rotation but keep the Gaia sampling.

On the test half, the 6,238 confident (tier A) periods are correct within 3% in 89.3% of cases (1303/1459), and in 95.4% (354/371) where the reference period has quality $U \geq 3-$. Many of the remaining disagreements are daily aliases in the ground-based reference periods. The spin sense agrees with published pole solutions in 98% of cases. I list 861 tier-A periods with no published value in the sources checked; their expected precision is about 90%, and null tests limit noise to at most 3.7%. The pipeline processes DR3 in about 20 minutes on ten cores and is meant to run on Gaia DR4 on release day.

Keywords: Asteroid rotation (2211) — Light curves (918) — Main belt asteroids (2036) — Photometry (1234)

1. INTRODUCTION

Gaia DR3 gave sparse photometry for about 157,000 numbered asteroids (Tanga et al. 2023). Three groups have derived rotation periods or spin states from these data, with different methods: Ďurech & Hanuš (2023), Cellino et al. (2024) and MacLennan et al. (2026). Many asteroids with enough transits still have no published period.

Two things are needed before such periods can be used:

1. an estimate of how often a period is wrong;
2. the same estimate for asteroids that are *not* already in other catalogues.

This paper provides both. It also makes the pipeline fast enough to rerun on Gaia DR4, expected in December 2026 with about twice the time baseline. A pilot on 600 asteroids was reported in Tropynin (2026).

2. DATA AND GEOMETRY

All 20 bulk files of `sso_observation` (23.3 M CCD rows) were averaged per `transit_id`. This gives 3,069,170 transits of 156,789 numbered asteroids. Of these, 42,195 asteroids have at least 25 transits. Each transit carries the largest relative flux error of its CCDs, with a 0.03 mag floor.

Heliocentric positions come from JPL osculating elements (epoch 2026 June). These were integrated back to 2014–2017 with a fourth-order Runge–Kutta scheme, a one-day step, and the Sun and giant planets as perturbers. Gaia's position is taken from the data, and light time is iterated. Against JPL Horizons on 19 asteroids, reduced magnitudes agree to 0.0007 mag.

3. MODEL AND SEARCH

The reduced magnitude $G - 5\log_{10}(r\Delta)$ is fitted by weighted least squares. The model has three parts:

- one constant per observing season;
- a term linear in phase angle;
- a second-order Fourier series in the phase $2\pi(ft - s\,\lambda_{\rm PAB}/360^\circ)$.

Here $\lambda_{\rm PAB}$ is the ecliptic longitude of the phase-angle bisector seen from Gaia, and $s = \pm 1$ is the sense of

rotation. Both senses are searched, and the better fit is kept.

Periods of 2–1000 h are scanned on a grid with step $0.5/T$, where T is the time span; the 30 deepest minima are then refined. A grid five times finer changed precision by less than a point. If the first-harmonic amplitude exceeds 1.20 times the second, the doubled period is adopted, because most lightcurves are double-peaked.

On the test set described below, the PAB correction had three effects:

- the fraction of correct periods rose from 45.5% to 52.6%;
- the median difference from the sidereal periods of Ďurech & Hanuš (2023) fell by a factor of four;
- the period dependence of that difference, expected for an uncorrected synodic drift, largely disappeared ($9\times10^{-5} \to 6\times10^{-6}$ at 2–5 h, $3\times10^{-4} \to 7\times10^{-5}$ at 10–30 h).

4. CALIBRATION AND NULL TESTS

The truth set is LCDB V4.0 (Warner et al. 2009) with quality $U \geq 2$. Its last update (September 2021) predates Gaia DR3. A period counts as correct if it lies within 3% of the LCDB value. The 9,613 known asteroids were split by a hash of their number into TUNE (4,779) and TEST (4,834).

On TUNE I fitted the doubling rule and a logistic model of the probability that a period is correct. The model uses seven features: the χ^2 gain over a no-rotation model, the separation from the next non-alias minimum, the number of transits, the amplitude, the reduced χ^2, the period and the harmonic ratio. On TEST, the predicted probabilities follow the observed fractions (Figure 2). The top bin is slightly overconfident: 0.95 predicted against 0.89 observed.

The threshold for confident periods was not chosen on TEST. I ran two null tests on 4,000 random asteroids without a known period:

- *white*: the residuals of the no-rotation model were permuted within seasons;
- *block*: one-day blocks of residuals were permuted, so that correlated systematics survive.

Noise periods match no catalogue, so they all end up in the uncatalogued subset. I therefore required, before choosing a threshold, that the projected noise be below 5% even in that subset.

At the TUNE threshold for 80% precision (0.384), 19.8% (white) and 18.4% (block) of signal-free asteroids pass. At 0.80 the pass rates are 0.10% and 0.05% (Figure 3). Tier A is therefore defined as probability ≥ 0.80 and period < 100 h. Periods of 100–1000 h are flagged as slow candidates and are never treated as confident.

5. RESULTS

5.1. *Periods*

Of 41,842 solutions, 6,238 are in tier A. On TEST, tier A is correct in 1303 of 1459 cases (89.3%): 90%, 88% and 83% in the 2–10, 10–30 and 30–100 h ranges (Figure 1). The result depends strongly on the quality of the reference period:

- 95.4% (354/371) where LCDB has $U \geq 3-$;
- 89.1% for $U = 2$ and 2+;
- 36% for $U = 2-$.

Of the 156 disagreements:

- 58 differ from LCDB by exactly one cycle per day in lightcurve frequency, the daily alias of single-site photometry;
- 6 differ by a factor of two;
- 92 are other periods.

Where an independent TESS period exists (Vavilov & Carry 2025), it sides with this work in 20 of these 38 cases and with LCDB in 3. Many apparent errors are therefore errors in the reference.

On TEST asteroids that appear in none of the six catalogues of Table 1, the closest known analogue of the unknown population, tier A is correct in 185 of 205 cases (90%).

The χ^2-curvature errors are too small. Against Ďurech & Hanuš (2023), the median $|\Delta P/P|$ is 3.9×10^{-5}. Adding $5.5\times10^{-5}P$ in quadrature gives a median $|\Delta P|/\sigma \approx 0.67$. This calibration measures agreement between two methods on the same data; it does not include aliasing.

5.2. *Spin senses*

The sense s chosen by the PAB fit agrees with the hemisphere of published poles in three comparisons:

- 3473 of 3535 asteroids (98.2%) in Ďurech & Hanuš (2023);
- 3802 of 3899 (97.5%) in DAMIT (Ďurech et al. 2010);
- 372 of 394 (94.4%) for DAMIT asteroids that are not in Ďurech & Hanuš (2023).

Table 1. Agreement of tier-A periods with published catalogues (within 3%).

Catalogue	Data	n	Agree
Ďurech & Hanuš (2023)	Gaia DR3	3564	99.6%
Cellino et al. (2024)	Gaia DR3	2883	99.0%
MacLennan et al. (2026)	Gaia DR3	2676	99.0%
Vavilov & Carry (2025)	TESS	759	91.0%
Erasmus et al. (2020)	ATLAS	280	68.6%
McNeill et al. (2023)	TESS	863	55.0%

NOTE—The Gaia DR3 rows test consistency, not accuracy. For TEST asteroids also in LCDB, the number of cases that agree with LCDB, this work vs. the catalogue: 373 vs. 359 of 411 (Vavilov & Carry), 111 vs. 116 of 146 (Erasmus et al.) and 352 vs. 231 of 379 (McNeill et al.).

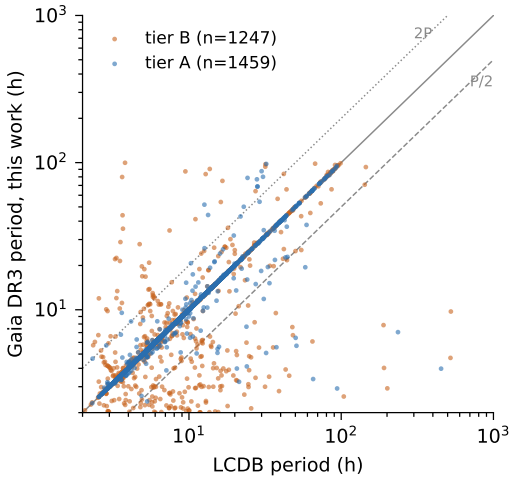


Figure 1. Period from this work against the LCDB period for TEST asteroids. Blue: tier A; orange: tier B. Lines mark P, $2P$ and $P/2$. The off-diagonal scatter includes daily aliases in the reference periods.

Poles within 20° of the ecliptic and ambiguous pole pairs were excluded. The catalogue gives the sense for every tier-A asteroid: 3,677 retrograde and 2,561 prograde, of which 3,390 have no LCDB period.

5.3. *New periods*

942 tier-A asteroids have a period neither in LCDB nor in the six catalogues. For 81 of them a rotation period appears elsewhere:

- in DAMIT;
- in about 180 VizieR tables with rotation periods (5.0 M rows), with units taken from the table metadata and non-rotation periods excluded;
- in the Minor Planet Bulletin, volumes 50–53.

Where values can be compared, the disagreements have known causes:

- partial-coverage PTF periods that are half of ours;
- K2 periods that differ by 3–6%;
- fast rotators below our 2 h limit;
- two slow rotators.

The remaining 861 asteroids (667 at 2–10 h, 131 at 10–30 h, 63 at 30–100 h) are listed as new. The matched control gives an expected precision of about 90%. The null tests allow at most 16–32 noise periods, or $\leq 3.7\%$ of the list. SsODNet, Behrend's Geneva page and ZTF period papers outside VizieR were not searched.

Of the 42 candidates of Tropynin (2026):

- 35 agree with this work;
- 4 flagged "low" have their period doubled here;
- 3 flagged "high" (39932, 109412, 13686) now prefer slow solutions outside tier A and should be treated as ambiguous.

6. LIMITATIONS AND PLANS FOR GAIA DR4

Periods below 2 h are not searched. The A_1/A_2 doubling rule is a heuristic. For doubled periods the PAB coefficient should be 2; a variant that also tries 2 improved TEST precision only from 89.3% to 89.7%. Tumblers fall outside a single-period model, and injection–recovery tests on the real Gaia sampling are still to be done.

DR4 will have about 66 months of data. For it I plan to fit the sidereal period and the pole jointly on a coarse pole grid, with a model-based comparison of P and $2P$. Thresholds and null tests will be fixed in advance, so that results can be released within a day.

7. DATA AVAILABILITY

The catalogue (41,842 rows, with spin senses), the list of 861 new periods, the validation tables and the code will be archived on Zenodo [DOI to be assigned].

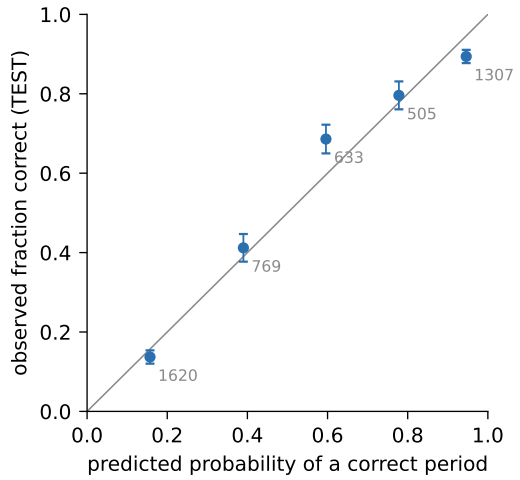


Figure 2. Reliability diagram on TEST, all solutions. Bars are 95% binomial intervals; numbers give the count per bin.

This work used data from the ESA mission Gaia, processed by the Gaia DPAC; JPL Horizons and the JPL Small-Body Database; LCDB; DAMIT; and VizieR (CDS, Strasbourg). The analysis was designed, coded and checked with the help of an AI assistant (Claude, Anthropic) under the author's direction. Two rounds of AI-based referee review were used to stress-test the validation.

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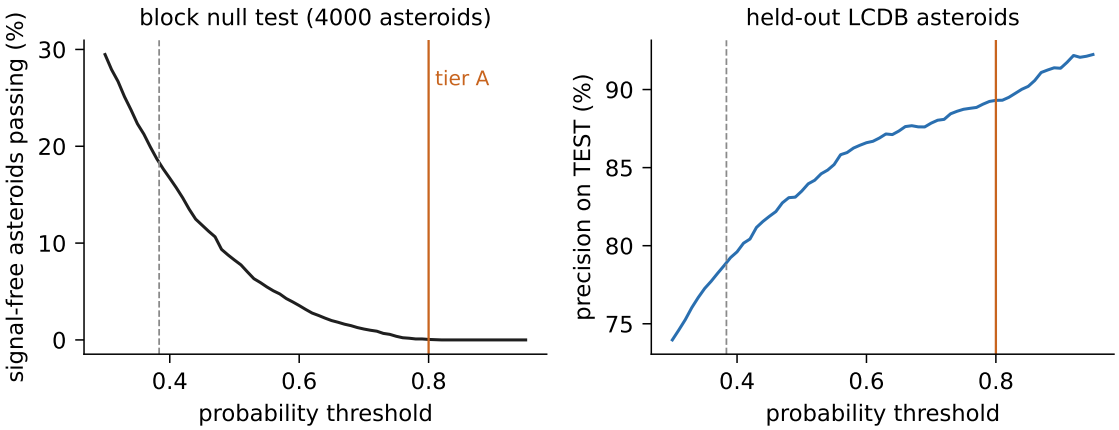


Figure 3. Left: fraction of signal-free asteroids (block null test) that pass, as a function of the probability threshold. Right: precision on TEST. Dashed line: the TUNE threshold for 80% precision. Solid line: the tier-A threshold.